# Homework 4: Standard Solutions
# 复分析第 4 次作业标准解答

Each English sentence is followed by its Chinese translation; a displayed formula is shared by the corresponding pair.
每个英文句子后紧接对应的中文译文；陈列公式由对应的两句共用。

The solutions use only Cauchy's integral formula and its elementary consequences, power series, and elementary real integration.
解答仅使用柯西积分公式及其初等推论、幂级数和初等实积分。

## 1. Generalized Liouville's theorem / 广义刘维尔定理

**Problem / 原题**

Prove the following *generalized Liouville's theorem*: If $f$ is an entire function and $|f(z)| \le C(1+|z|^N)$ for all $z \in \mathbb{C}$, then $f$ is a polynomial of degree at most $N$.
证明下述广义刘维尔定理：若 $f$ 是整函数，且对所有 $z \in \mathbb{C}$ 都有 $|f(z)| \le C(1+|z|^N)$，则 $f$ 是次数不超过 $N$ 的多项式。

**Solution / 解答**

As usual, $C$ is a fixed positive constant and $N$ is a nonnegative integer.
按通常约定，$C$ 是固定的正常数，$N$ 是非负整数。

Write the Taylor expansion of the entire function at the origin as follows.
将整函数在原点的泰勒展开写为如下形式。

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \qquad z \in \mathbb{C}.$$

For every $R > 0$, Cauchy's coefficient formula and the given bound imply the following estimate.
对任意 $R > 0$，由柯西系数公式及题设的上界可得如下估计。

$$a_n = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{f(\zeta)}{\zeta^{n+1}} \, \mathrm{d}\zeta, \qquad |a_n| \le \frac{\max_{|\zeta|=R} |f(\zeta)|}{R^n} \le C\big(R^{-n} + R^{N-n}\big).$$

Fix an integer $n > N$ and let $R \to \infty$; both powers on the right tend to zero, so $a_n = 0$.
固定整数 $n > N$ 并令 $R \to \infty$；右端的两个幂次项都趋于零，故 $a_n = 0$。

Consequently, $f$ has the following form and is a polynomial of degree at most $N$, with the zero polynomial included.
因此，$f$ 具有如下形式，是次数不超过 $N$ 的多项式，其中包括零多项式。

$$\boxed{f(z) = \sum_{n=0}^{N} a_n z^n.}$$

## 2. Cartan's uniqueness theorem / Cartan 唯一性定理

**Problem / 原题**

Prove the following *Cartan's uniqueness theorem*.

证明下述 *Cartan* 唯一性定理。

Let $f$ be a holomorphic self-mapping of a bounded domain $\Omega$.

设 $f$ 是有界区域 $\Omega$ 的全纯自映射。

Suppose that there exists $z_0 \in \Omega$ such that $f(z_0) = z_0$ and $f'(z_0) = 1$.

假设存在 $z_0 \in \Omega$，使得 $f(z_0) = z_0$ 且 $f'(z_0) = 1$。

Then $f$ is the identity.

则 $f$ 是恒等映射。

*Hint.* Without loss of generality, suppose that $z_0 = 0$.

提示。不妨设 $z_0 = 0$。

Assume that $f(z) = z + a_k z^k + O(z^{k+1})$ with $a_k \neq 0$ for some $k \geq 2$.

假设对某个 $k \geq 2$，有 $f(z) = z + a_k z^k + O(z^{k+1})$，其中 $a_k \neq 0$。

Denote by $f^n$ the $n$-th iterate of $f$.

用 $f^n$ 表示 $f$ 的第 $n$ 次迭代。

Apply the Cauchy estimates to $f^n$ to get a contradiction.

对 $f^n$ 应用柯西估计以得到矛盾。

**Solution / 解答**

Replacing $\Omega$ by $\Omega - z_0$ and $f$ by $w \mapsto f(w + z_0) - z_0$, we may assume that $0 \in \Omega$, $f(0) = 0$, and $f'(0) = 1$.

将 $\Omega$ 换为 $\Omega - z_0$，将 $f$ 换为 $w \mapsto f(w + z_0) - z_0$，便可假设 $0 \in \Omega$、$f(0) = 0$ 且 $f'(0) = 1$。

Suppose, for a contradiction, that $f$ is not the identity on $\Omega$.

为导出矛盾，假设 $f$ 在 $\Omega$ 上不是恒等映射。

By the Taylor expansion and the identity theorem, there is a least integer $k \geq 2$ with a nonzero coefficient in $f(z) - z$, giving the following local expansion.

由泰勒展开和恒等定理，$f(z) - z$ 中存在一个系数非零的最低次项，其次数为某个 $k \geq 2$，从而有如下局部展开。

$$f(z) = z + a_k z^k + O(z^{k+1}), \qquad a_k \neq 0.$$

Let $F_n = f^{\circ n}$ denote the $n$-fold composition; we claim that its expansion at zero is as follows.

用 $F_n = f^{\circ n}$ 表示 $f$ 的 $n$ 次复合；我们断言它在零点处具有如下展开。

$$F_n(z) = z + n a_k z^k + O(z^{k+1}), \qquad n \geq 1.$$

The case $n = 1$ is immediate.

当 $n = 1$ 时，断言直接成立。

If the claim holds for $n$, then $F_n(z) = z + O(z^k)$ and $k \geq 2$ imply the following relations.

若断言对 $n$ 成立，则由 $F_n(z) = z + O(z^k)$ 及 $k \geq 2$ 得到下列关系。

$$F_n(z)^k = z^k + O(z^{2k-1}) = z^k + O(z^{k+1}), \qquad F_n(z)^{k+1} = O(z^{k+1}).$$

Substitution into the expansion of $f$ proves the induction step.
将其代入 $f$ 的展开，便完成归纳步骤。

$$\begin{aligned} F_{n+1}(z) &= f(F_n(z)) \\ &= F_n(z) + a_k F_n(z)^k + O(F_n(z)^{k+1}) \\ &= z + (n+1)a_k z^k + O(z^{k+1}). \end{aligned}$$

Choose $r > 0$ such that $\overline{D(0;r)} \subset \Omega$, and choose $M > 0$ such that $|w| \leq M$ for every $w \in \Omega$.
选取 $r > 0$ 使得 $\overline{D(0;r)} \subset \Omega$，并选取 $M > 0$ 使得对每个 $w \in \Omega$ 都有 $|w| \leq M$。
Since $f$ maps $\Omega$ into itself, every $F_n$ is holomorphic on $\Omega$ and satisfies $|F_n(z)| \leq M$ there.
由于 $f$ 将 $\Omega$ 映入自身，每个 $F_n$ 都在 $\Omega$ 上全纯，且在其中满足 $|F_n(z)| \leq M$。
Cauchy's estimate for the coefficient of $z^k$ in the expansion of $F_n$ therefore yields a bound independent of $n$.
因此，对 $F_n$ 的展开中 $z^k$ 的系数应用柯西估计，得到一个与 $n$ 无关的上界。

$$n|a_k| = \left| \frac{F_n^{(k)}(0)}{k!} \right| \leq \frac{M}{r^k}, \qquad n \geq 1.$$

Letting $n \to \infty$ contradicts $a_k \neq 0$, so the translated map is the identity, and translating back gives $f(z) = z$ throughout the original domain.
令 $n \to \infty$，便与 $a_k \neq 0$ 矛盾，故平移后的映射为恒等映射，平移回去即得原区域上处处有 $f(z) = z$。

## 3. Gutzmer's inequality / Gutzmer 不等式

**Problem / 原题**

Let $f(z)=\sum_{n=0}^{\infty}a_nz^n$ be a power series with radius of convergence $\rho>0$.
设 $f(z)=\sum_{n=0}^{\infty}a_nz^n$ 是收敛半径为 $\rho>0$ 的幂级数。
Prove the following *Gutzmer's inequality*:
证明如下 *Gutzmer* 不等式：

$$\sum_{n=0}^{\infty}|a_n|^2r^{2n}\le A_f(r)^2,\qquad 0<r<\rho.$$

where $A_f(r):=\max\{|f(z)|:|z|=r\}$.
其中 $A_f(r):=\max\{|f(z)|:|z|=r\}$。
*Hint.* Consider the integral $\frac{1}{2\pi}\int_0^{2\pi}|f(re^{i\theta})|^2\,\mathrm{d}\theta$.
提示。考虑积分 $\frac{1}{2\pi}\int_0^{2\pi}|f(re^{i\theta})|^2\,\mathrm{d}\theta$。

**Solution / 解答**

Fix $0<r<\rho$ and define the partial sums on the circle by the following formula.
固定 $0<r<\rho$，按下式定义圆周上的部分和。

$$S_m(\theta)=\sum_{n=0}^{m}a_nr^ne^{in\theta},\qquad 0\le\theta\le 2\pi.$$

The power series converges uniformly on $|z|=r$, so $S_m(\theta)\to f(re^{i\theta})$ uniformly in $\theta$.
幂级数在 $|z|=r$ 上一致收敛，因此 $S_m(\theta)$ 关于 $\theta$ 一致收敛到 $f(re^{i\theta})$。
The elementary orthogonality identity below evaluates the integral of each finite sum.
下面的初等正交恒等式可以计算每个有限和的积分。

$$\frac{1}{2\pi}\int_0^{2\pi}e^{i(n-j)\theta}\,\mathrm{d}\theta=\begin{cases}1, & n=j,\\ 0, & n\ne j.\end{cases}$$

Expanding the squared modulus and integrating term by term in this finite sum gives the following equality.
展开模的平方，并对这个有限和逐项积分，得到如下等式。

$$\begin{aligned}\frac{1}{2\pi}\int_0^{2\pi}|S_m(\theta)|^2\,\mathrm{d}\theta&=\sum_{n,j=0}^{m}a_n\overline{a_j}\,r^{n+j}\frac{1}{2\pi}\int_0^{2\pi}e^{i(n-j)\theta}\,\mathrm{d}\theta\\&=\sum_{n=0}^{m}|a_n|^2r^{2n}.\end{aligned}$$

Uniform convergence also gives uniform convergence of the squared moduli, by the following estimate.
由下面的估计可知，一致收敛还蕴含模的平方一致收敛。

$$\big||S_m(\theta)|^2-|f(re^{i\theta})|^2\big|\le|S_m(\theta)-f(re^{i\theta})|\big(|S_m(\theta)|+|f(re^{i\theta})|\big).$$

Passing to the limit as $m \to \infty$ therefore proves the desired inequality, together with an exact integral identity.

因此，令 $m \to \infty$，便得到所需不等式，同时得到一个精确的积分恒等式。

$$\boxed{\sum_{n=0}^{\infty} |a_n|^2 r^{2n} = \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 \, \mathrm{d}\theta \leq A_f(r)^2.}$$

# 4. An integral on the unit circle / 单位圆周上的积分

**Problem / 原题**

Compute

$$\oint_{|z|=1} \frac{|\mathrm{d}z|}{|z-a|^2}, \qquad 0<|a|<1.$$

by transforming the integral into a form where you can apply the Cauchy integral formula.
将上式中的积分化为可以应用柯西积分公式的形式，并计算其值。

**Solution / 解答**

The integral is with respect to arc length, so we may choose the counterclockwise parametrization $z=e^{it}$, $0\le t\le 2\pi$.
该积分关于弧长进行，故可选取逆时针参数化 $z=e^{it}$，其中 $0\le t\le 2\pi$。

On this parametrized circle, the following identities hold.
在这样参数化的圆周上，有下列恒等式。

$$\overline{z}=\frac{1}{z}, \qquad \mathrm{d}z=iz\,\mathrm{d}t, \qquad |\mathrm{d}z|=\mathrm{d}t=\frac{\mathrm{d}z}{iz},$$
$$|z-a|^2=(z-a)(\overline{z}-\overline{a})=\frac{(z-a)(1-\overline{a}z)}{z}.$$

Substitution transforms the arc length integral into the following complex contour integral, taken counterclockwise.
代入上述恒等式，就将弧长积分化为下面沿逆时针方向进行的复曲线积分。

$$I:=\oint_{|z|=1}\frac{|\mathrm{d}z|}{|z-a|^2}=\frac{1}{i}\oint_{|z|=1}\frac{1}{(z-a)(1-\overline{a}z)}\,\mathrm{d}z.$$

Set $g(z)=1/(1-\overline{a}z)$; since its only possible singularity is at $1/\overline{a}$, whose modulus is greater than 1, $g$ is holomorphic on a neighborhood of the closed unit disk.
令 $g(z)=1/(1-\overline{a}z)$；由于它唯一可能的奇点 $1/\overline{a}$ 的模大于 1，故 $g$ 在闭单位圆盘的某个邻域上全纯。

Because $|a|<1$, Cauchy's integral formula applies directly at $a$ and gives the result.
由于 $|a|<1$，可在 $a$ 处直接应用柯西积分公式，得到结果。

$$I=\frac{1}{i}\oint_{|z|=1}\frac{g(z)}{z-a}\,\mathrm{d}z=\frac{1}{i}\,2\pi i\,g(a)=\boxed{\frac{2\pi}{1-|a|^2}}.$$

# 5. The area mean value property / 面积平均值性质

**Problem / 原题**

Prove the following version of the *mean value property*: Let $\Omega$ be a domain and $f \in A(\Omega)$.

证明下述形式的平均值性质：设 $\Omega$ 是一个区域，且 $f \in A(\Omega)$。

Suppose that $D(z_0;\rho) \subset \Omega$.

假设 $D(z_0;\rho) \subset \Omega$。

Then the following equality holds.

则下列等式成立。

$$f(z_0) = \frac{1}{\pi\rho^2} \iint_{D(z_0;\rho)} f(z)\,\mathrm{d}\lambda(z).$$

**Clarification of the hypothesis / 题设说明**

Here $A(\Omega)$ is understood to mean the holomorphic functions on $\Omega$, and $\lambda$ denotes planar area measure.

这里将 $A(\Omega)$ 理解为 $\Omega$ 上的全纯函数空间，$\lambda$ 表示平面面积测度。

If $D(z_0;\rho)$ denotes an open disk, the stated inclusion alone does not guarantee that $f$ is integrable over that disk.

若 $D(z_0;\rho)$ 表示开圆盘，则题中给出的包含关系本身不能保证 $f$ 在该圆盘上可积。

For an ordinary area integral, a sufficient additional condition is $\overline{D(z_0;\rho)} \subset \Omega$, or, more generally, $f \in L^1(D(z_0;\rho))$.

若采用通常的面积积分，一个充分的补充条件是 $\overline{D(z_0;\rho)} \subset \Omega$，或者更一般地，$f \in L^1(D(z_0;\rho))$。

If the notation $D$ in the course already denotes a closed disk, its stated inclusion supplies the sufficient condition directly.

若课程中的记号 $D$ 本来就表示闭圆盘，则原题中的包含关系已直接给出上述充分条件。

**Solution under the usual integrability condition / 在通常可积条件下的解答**

First fix $0 < R < \rho$; its closed disk is contained in $\Omega$, so all integrals below over this smaller disk are well defined.

先固定 $0 < R < \rho$；相应闭圆盘包含于 $\Omega$，因而下面在这个较小圆盘上的所有积分都有定义。

For every $0 < r \le R$, Cauchy's integral formula and the parametrization $\zeta = z_0 + re^{it}$ give the circle mean value identity.

对每个 $0 < r \le R$，由柯西积分公式及参数化 $\zeta = z_0 + re^{it}$ 得到圆周平均值恒等式。

$$f(z_0) = \frac{1}{2\pi i} \oint_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z_0}\,\mathrm{d}\zeta = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it})\,\mathrm{d}t.$$

Using polar coordinates about $z_0$, whose area element is $r\,\mathrm{d}r\,\mathrm{d}t$, we obtain the following equality.

采用以 $z_0$ 为中心的极坐标，其面积元为 $r\,\mathrm{d}r\,\mathrm{d}t$，可得如下等式。

$$\begin{aligned} \iint_{D(z_0;R)} f(z)\,\mathrm{d}\lambda(z) &= \int_0^R \left( \int_0^{2\pi} f(z_0 + re^{it})\,\mathrm{d}t \right) r\,\mathrm{d}r \\ &= \int_0^R 2\pi f(z_0)\, r\,\mathrm{d}r = \pi R^2 f(z_0). \end{aligned}$$

If $f \in L^1(D(z_0;\rho))$, letting $R \uparrow \rho$ is justified by integrability, and yields the required formula.
若 $f \in L^1(D(z_0;\rho))$，则可由可积性令 $R \uparrow \rho$，得到所需公式。

$$\boxed{\frac{1}{\pi\rho^2}\iint_{D(z_0;\rho)} f(z)\,\mathrm{d}\lambda(z) = f(z_0).}$$

Indeed, the absolute value of the difference between the full integral and the integral over $D(z_0;R)$ is at most the integral of $|f|$ over the remaining annulus, which tends to zero.
事实上，全圆盘积分与 $D(z_0;R)$ 上积分之差的绝对值，不超过剩余圆环上 $|f|$ 的积分，而后者趋于零。

In particular, when $\overline{D(z_0;\rho)} \subset \Omega$, continuity makes $f$ bounded on that closed disk, so this integrability condition is automatic.
特别地，当 $\overline{D(z_0;\rho)} \subset \Omega$ 时，连续性保证 $f$ 在该闭圆盘上有界，因而上述可积性条件自动成立。

**Why the open disk hypothesis needs care / 为何开圆盘条件需要说明**

For a counterexample to automatic integrability, take $\Omega = D(0;1)$ and $f(z) = (1-z)^{-2}$.
为说明可积性并不自动成立，取 $\Omega = D(0;1)$ 及 $f(z) = (1-z)^{-2}$。

This function is holomorphic on $\Omega$, but its absolute area integral diverges.
该函数在 $\Omega$ 上全纯，但其绝对值的面积积分发散。

To see this, write $z = 1 - se^{i\theta}$ and restrict to $0 < s < 1/2$ and $|\theta| < \pi/3$; this sector lies inside $D(0;1)$ because of the following inequality.
为验证这一点，写成 $z = 1 - se^{i\theta}$ 并限制 $0 < s < 1/2$、$|\theta| < \pi/3$；由下列不等式可知，该扇形包含于 $D(0;1)$。

$$|z|^2 = 1 - 2s\cos\theta + s^2 \le 1 - s + s^2 < 1.$$

On this sector, $|f(z)| = s^{-2}$ and the area element is $s\,\mathrm{d}s\,\mathrm{d}\theta$, so the following divergence holds.
在该扇形上，$|f(z)| = s^{-2}$，面积元为 $s\,\mathrm{d}s\,\mathrm{d}\theta$，因此有如下发散关系。

$$\iint_{D(0;1)} |f(z)|\,\mathrm{d}\lambda(z) \ge \int_{-\pi/3}^{\pi/3}\int_0^{1/2} \frac{1}{s^2}\, s\,\mathrm{d}s\,\mathrm{d}\theta = \infty.$$

Thus the ordinary complex Lebesgue integral in the original statement need not exist if only the open disk inclusion is assumed.
因此，若只假设开圆盘包含关系，原题中的通常复值勒贝格积分可能不存在。

Without any extra condition, the result remains valid as an improper integral defined by concentric disks, since the identity for $R < \rho$ already gives the following limit.
无须任何额外条件，该结论仍可理解为通过同心圆盘定义的反常积分，因为 $R < \rho$ 时的恒等式已经给出如下极限。

$$\lim_{R\uparrow\rho} \iint_{D(z_0;R)} f(z)\,\mathrm{d}\lambda(z) = \pi\rho^2 f(z_0).$$